

Conway Garden Control Box

Printable wiring overview for client review. This keeps the diagram readable and uses WireViz as the source and BOM artifact instead of making the long WireViz render the main client view.

Revision 0.3

2026-07-08

WireViz source backed

Power entry

120 V AC three-prong input enters through the fused inlet and breaker.

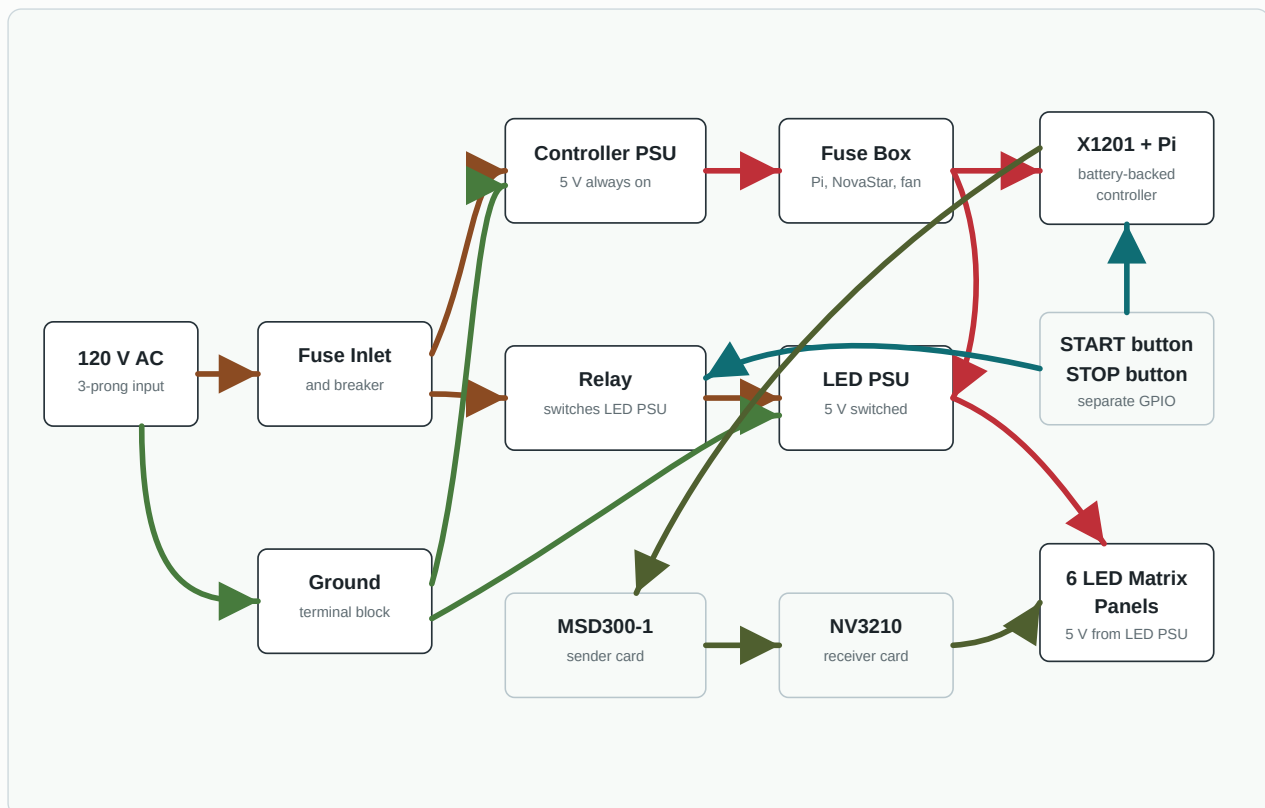
Two 5 V supplies

The controller PSU stays on. The LED-panel PSU is switched by the relay.

Display chain

Pi micro HDMI to DVI, MSD300-1 sender, one Ethernet cable, NV3210, ribbon cables, six panels.

Compact Wiring Diagram



 AC line/neutral	 Ground
 5 V DC	 Button/relay control
 Video/data	

What Turns Off

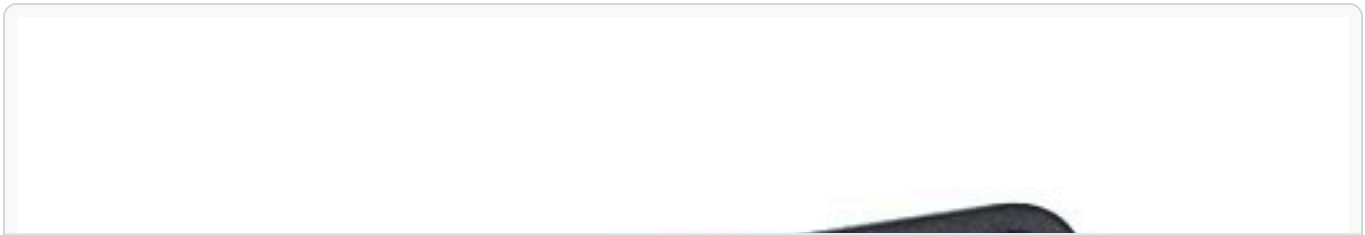
- The Pi controller path stays powered by the controller 5 V PSU and X1201 battery backup.
- The relay switches the LED-panel PSU only.
- When the relay is off, the six LED panels lose 5 V panel power while the Pi remains available.

Protected Outputs

- The fuse box protects the X1201/Pi branch.
- The fuse box protects the NovaStar MSD300-1 sender.
- The fuse box protects the NovaStar NV3210 receiver and the enclosure fan.
- The fuse box does not protect or feed the LED panels.

This packet is for client review and documentation. Final energized work still needs fuse ratings, wire gauge, strain relief, grounding, terminal torque, relay load/heat, and local electrical-code review.

Component Photos



Power inlet and breaker

Protected entry point for the 120 V AC three-prong cable.

Photo source



Ground terminal block

Case-bonded ground point that connects ground to both power supplies.

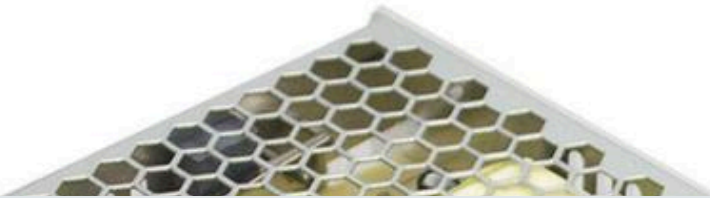
Photo source



Controller 5 V PSU

Always-on MEAN WELL LRS-50-5 supply for Pi-side electronics.

Photo source

**LED panel 5 V PSU**

Relay-switched MEAN WELL supply for the LED matrix panels.

Photo source

**Display power relay**

Pi-controlled relay that switches only the LED-panel PSU line input.

Photo source

**Fuse box**

Protects X1201/Pi, MSD300-1, NV3210, and fan branches.

Photo source

Geekworm X1201

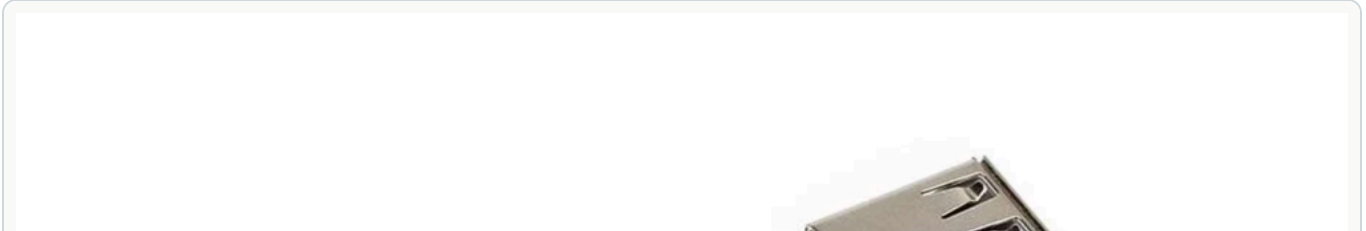
UPS HAT that keeps the Raspberry Pi alive during power loss.

Photo source

**Samsung 30Q cells**

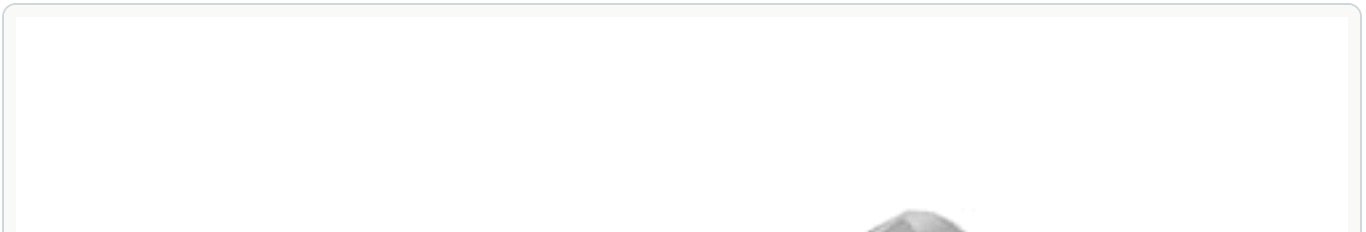
Two 3.7 V 18650 cells for the X1201 backup board.

Photo source

**Raspberry Pi 5**

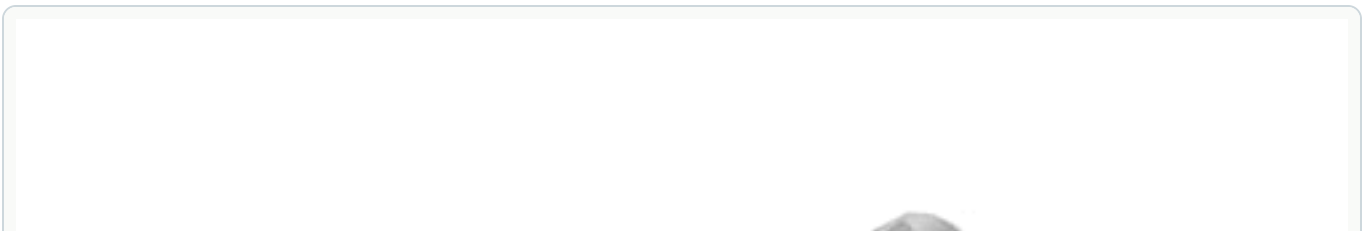
Runs the garden controller, video output, buttons, and relay control.

Photo source

**START button**

Dedicated front-panel START input for the controller.

Photo source

**STOP button**

Dedicated front-panel STOP input. It stops the display, not Pi power.

Photo source

Micro HDMI to DVI

Adapts Raspberry Pi video output for the MSD300-1 sender.

Photo source

**NovaStar MSD300-1**

Receives DVI video and sends display data over Ethernet.

Photo source

**Ethernet cable**

One complete cable from the MSD300-1 sender to the NV3210 receiver.

Photo source

**NovaStar NV3210**

Receiver card that connects to the LED matrix panels by ribbon cable.

Photo source



LED matrix panels (x6)

Six panels receive 5 V from the LED PSU and data from the NV3210.

Photo source



Enclosure fan

Small 5 V fan on the protected controller/display-electronics side.

Photo source

WireViz Appendix

WireViz remains the editable source of record for the simplified wiring model and generated BOM. The long raw render is useful as an appendix, not as the primary client-facing page.

WireViz source YAML

WireViz HTML render and BOM

WireViz SVG render

WireViz BOM TSV

Source Notes

- START button is GPIO18. STOP button is GPIO19. Relay control is GPIO20.
- The X1201 PLD signal remains the Pi-side power-loss input.
- The LED-panel PSU is separate from the controller PSU so the display can be switched without cutting Pi power.
- The display path is Raspberry Pi micro HDMI, DVI into MSD300-1, one Ethernet cable to NV3210, then ribbon cables to six LED matrix panels.

Generated PDF target: out/conway-garden-client-print.pdf